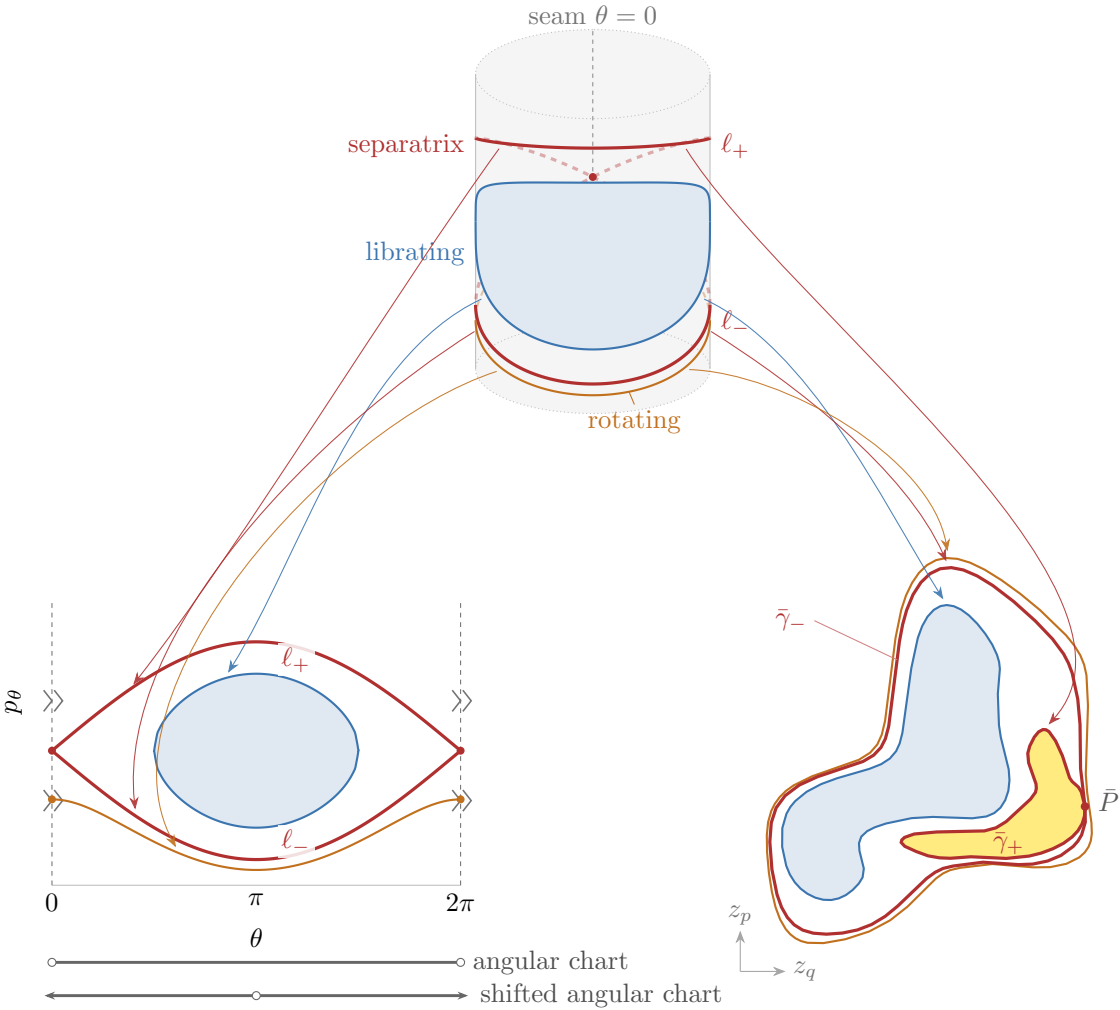

seam $\theta = 0$
separatrix
$\ell_+$
librating
$\ell_-$
rotating
$\bar{\gamma}_-$
$\bar{P}$
$\bar{\gamma}_+$
$z_p$
$z_q$
$p_\theta$
$\ell_+$
$\ell_-$
0
$\pi$
$2\pi$
$\theta$
angular chart
shifted angular chart
(i)
(ii)